MILITARY SPECIFICATION

WIND MEASURING SET AN/UMQ-5 ( )
(Components of)

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers the following components of Wind Measuring Set AN/UMQ-5 ( ):

a. Support MT-535( )/UMQ-5

b. Wind Direction-velocity Indicator ID-300( )/UMQ-5A

c Recorder, Wind Direction and Speed RD-108 ( )/UMQ-5

d. Wind Direction-velocity Transmitter ML-400 ( )/UMQ-5

e Indicator, Wind Direction and Speed ID-586 ( )/UMQ-5

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-489	Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces)

FSC 6660

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Federal (Continued)

PPP-B-636	Box, Fiberboard

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-M-16034	Meter, Electrical Indicating (Switchboard and Portable Types)
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-20708	Synchros, General Specification for
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment

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Military (Continued)

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

DRAWINGS

Naval Air Systems Command

Drawing Set No. 58A43	Wind Measuring Set AN/UMQ-5C (Microfilm) (See Note 6.3)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The wind measuring set furnished under this specification shall be a propeller type housed in a streamlined vane mounted on a support, generating d. c. voltages for wind speed and utilizing a synchro for wind direction. Any movement of the vane and/or propeller shall be transmitted electrically to the indicators and recorders.

3.2 First article sample. A first article sample or samples as specified in the contract are required under this specification. Insofar as practicable, the manufacturer of first article samples shall utilize tools, methods and processes which will be used in full-scale production. The samples shall be approved by the requiring activity prior to manufacture of any production units. If required by the procuring agency, the first article sample shall be shipped, at no cost to the Government, to a testing center designated for tests and checks deemed necessary. If required by the procuring agency, the first article model shall be corrected and resubmitted for approval following any first article tests until finally approved. The finally approved model, unless destroyed or damaged beyond practical reconstruction by first article testing, shall be retained on hand for use by the inspector as necessary during subsequent production testing. Prior to completion of the contract, the first article model shall be corrected or reconstructed as necessary to become the equivalent of a production item and delivered with the last unit of production equipment. It shall not be counted as one of the production units.

3.3 General requirements.

3.3.1 Materials, parts, and processes. The wind equipment shall

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be in accordance with the following paragraphs of MIL-E-5400 in addition to the requirements specified herein. Whenever a requirement of MIL-E-5400 conflicts with a requirement of this specification, the requirement of this specification shall govern:

	Paragraph No.
Applicable Documents	2
Reordered Production Equipment	3.1.1.1.3
Choice of Parts and Materials	3.1.1.3
Equipment Performance	3.1.1.5
Castings	3.1.3
Controls	3.1.5
Corrosion Resistance	3.1.6
Dissimilar Metals	3.1.7
Finishes	3.1.10
Aluminum Alloy	3.1.10.1
Cadmium - Plated Parts	3.1.10.2
Zinc and Zinc-Plated Parts	3.1.10.7
Fungus - Inert Materials	3.1.11
Marking	3.1.16
Reference Designations	3.1.16.4
External Voltmeter Resistors	3.1.19.1
Soldering	3.1.20
Mechanical Assemblies	3.1.20.1
Springs	3.1.21
Toggle Switches	3.1.22.2
Terminals	3.1.23
Threaded Parts	3.1.24
Tools (Special)	3.1.25
Set Screw Wrenches	3.1.26
Wire (Hook-up)	3.1.32
Wiring (Internal)	3.1.33
Wood	3.1.35
Design and Construction	3.2
Thermal Design	3.2.5
Safety	3.2.22
Interchangeability	3.3
Workmanship	3.6
Notes	6

3.3.2 Service condition requirements. Wind equipment procured under this specification may be operated or stored under any possible combination of the service conditions as indicated below:

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a. Wind speed of 120 knots

b. Temperatures in the range -30°F to +160°F. Storage

c. Temperatures in the range -10°F to +130°F. Operating

d. Relative humidity up to 100%, including condensation caused by temperature changes

e. Salt-laden air – as encountered at coastal shore installations

f. Wind-driven rain at any velocity

g. Vibration and shock – as encountered in transportation

h. Fungi and insects – as encountered in tropical services

i. Exposure to sand and dust particles – as encountered in desert areas

3.3.2.1 The government reserves the right to require environmental testing as may be necessary to determine that equipment is capable of extended exposure, operation or storage under the above conditions. The contractor shall provide either directly in his own plant or through the facilities of an acceptable commercial laboratory all facilities for performing such environmental tests, including test equipment, a suitable location for the performance of tests and qualified technical personnel, unless testing at a government laboratory is required, in which case tests will be performed at government expense.

3.3.2.2 In lieu of environmental tests as above, at the option of the contracting officer, equipment may be accepted on the basis of a written guarantee by the contractor that he will replace, at no cost to the government, all units of equipment which fail, due to above causes during the one-year period commencing at the time the equipment is first placed into service use, and take such further corrective action as may be necessary to correct all units procured under the contract to prevent further failure due to these causes, such corrective action to be at no expense to the government, other than cost of shipment to and from the contractor's plant.

3.3.3 Design and construction. Design and construction shall conform to Naval Air Systems Command Drawing Set No. 58A43 except as specified herein, however, the manufacturer is responsible for furnishing a complete, fully functional end equipment, whether or not all required details are covered by these drawings. Such equipment is to be in compliance with all other requirements of this specification and applicable portions of referenced specifications. In the event of any conflicting requirements of the drawings and any other requirements of this

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specification, the specification requirements shall take precedence. The Naval Air Systems Command, shall, however, be advised in all instances where any such conflict appears to exist and Naval Air Systems Command approval shall be obtained prior to incorporating any deviations from the drawings into the equipment.

3.3.4 Wind set accuracy. The wind speed transmitter and wind speed indicators, and/or recorder, together shall be accurate to plus or minus 1-1/2 knots in the range of 3 to 40 knots and plus or minus 3 knots in the range of 41 to 120 knots. The vane shall align the transmitter within 3 degrees of true wind direction when displaced 10 degrees from the true wind direction in a 5 knot wind.

3.3.5 Wind measuring system. The wind measuring system shall provide for the operation from a single transmitter of from one to six indicators; or one to six recorders or any combination of indicators and recorders totaling not more than six units.

3.3.6 Support. The support shall be easily assembled and disassembled and so constructed as to enable the lowering of the support under normal conditions by one man. The support shall provide for convenient mounting of the transmitter and shall serve as a housing for the cable from the transmitter. Provisions shall be incorporated for the orientation of the support with respect to compass direction and the use of guys, if necessary.

3.3.7 Finish. Exterior metal components, except Support MT-535( )/UMQ-5, shall be thoroughly cleaned and painted as required with one coat of zinc-chromate primer conforming to MIL-P-8585 followed by one coat of gray enamel conforming to Class B of TT-E-489, Color No. 16473 per FED-STD-595. Support MT-535( )/UMQ-5 shall be hot-dip galvanized or zinc plated.

3.3.8 Nomenclature and nameplates. Nomenclature and designators used in paragraph 1.1 and elsewhere in this specification are for general reference use only and do not constitute specific assignments under any particular contract or order. Normally the parentheses will be replaced by a letter designating a specific model upon receipt and approval of the data required below. All drawings and nameplates required by this specification and the applicable contract or order shall bear only the specifically approved nomenclature and designators provided for below.

3.3.8.1 As soon as possible after award of contract and not later than 45 days thereafter, the contractor shall submit all data required below. If data cannot be furnished at this time, a written statement shall be promptly forwarded to the procuring agency indicating when such data can be furnished. Such statement shall not relieve the contractor of any contractual responsibility for delivery of material within the stipulated time.

3.3.8.2 MIL-N-18307 shall be used as a guide in the preparation and submission of necessary data and nameplates.

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3.4 Detailed requirements.

3.4.1 Support MT-535( )/UMQ-5. The support shall be 13 feet ±1 foot in length and shall have a three-legged tripod type base with a mounting diameter of 52 ±5 inches.

3.4.2 Wind direction-velocity indicator ID-300( )/UMQ-5A. The indicator shall contain, in an enclosure designed for desk use, two variable illuminated dials; one a direction indicator, on the left side of the panel, incorporating a Type 23TR6a Synchro, MIL-S-20708, for direction indications, and the other a speed indicator, on the right side, incorporating a high impact d.c. voltmeter, MIL-M-16034, for speed indications. The speed and direction indicators shall be mounted on a removable front panel of the enclosure, such that the indicator panel may be separately mounted in a standard 19-inch relay rack when desired.

3.4.2.1 Dials. The dials shall be 4 ±1/2 inches in diameter, and shall be dull black with white printing and graduations. The direction dial shall be graduated in 5-degree intervals with appropriate letters to designate the 4 cardinal and 4 inter-cardinal points of the compass. The speed dial shall contain two (2) scales, a lower scale, graduated from 0 to 60 knots in units of 1 knot, with distinctive marks at each 5 knots and with identification numbers at each 10 knots, and an upper scale, graduated from 0 to 120 knots in units of 2 knots, with distinctive marks at each 10 knots and with identification numbers at each 20 knots. The wind speed scale shall be graduated in 240 ±10 degrees of arc. White numerals and markings shall be semi-gloss white, Color No. 27875 per FED-STD-595.

3.4.2.2 Wind speed indicator. The wind speed voltmeter shall be linearly calibrated so that 7.29 volts d.c. equals 60 knots, have an accuracy of ±1 per cent, an internal resistance of 6,900 ohms ±1/2 per cent and perceptible movement to a voltage change of 0.090 volts. A range doubling toggle switch and a pilot light, indicating high speed range, shall be provided. A permanently mounted 6,900 ohm shunt resistor, tapped at values of 1,380, 1,725, 2,300, and 3,450 ohms, shall be provided as a convenient means to maintain the correct load on the transmitter magneto as the number of repeaters in the circuit is varied.

3.4.3 Recorder, wind direction and speed RD-108( )/UMQ-5. The recorder shall contain, in an enclosure designed for both desk and three-point suspension bulkhead mounting, a mechanism to record both wind direction and speed on a single illuminated strip chart, of which 6 inches of past record is always visible through the recorder door.

3.4.3.1 Chart. The wind direction section of the chart shall be graduated in six 90° quadrants as follows: left to right, N, E, S, W, N, E, S. The wind speed section shall be graduated from 0 to 120 knots. The chart shall be 100 feet long and have remaining footage marked at one foot intervals.

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3.4.3.2 Chart drive. The chart drive mechanism shall be powered by a totally enclosed 115 volt, 60 hertz, self-starting, synchronous motor with a built-in gear reduction unit, which together with two pairs of change gears shall be capable of furnishing chart speeds of 1-1/2, 3, and 6 inches per hour at an accuracy of one minute per 24 hours from a 60 hertz source. The chart feed rate will be normally 3 inches per hour with change gears provided to change speeds to 1-1/2 or 6 inches per hour without the use of tools. The chart drive mechanism shall contain a friction clutch or other means in the drive mechanism to permit the setting of the chart accurately to time. A stall torque electric motor driven take-up roll shall be provided to permit examination and reroll of the past record without interfering with the operation of the recorder.

3.4.3.3 Wind direction component. The wind direction component shall contain a mechanism, incorporating a Type 23TR6a Synchro, MIL-S-20708, for providing a curvilinear movement to a pen to provide a single trace of wind direction on the chart. The trace shall not require recourse to any identifying secondary pen marking. Automatic repositioning of the pen through 360 chart degrees shall be provided when the pen reaches either extremity of the chart. An indicator scale shall be provided which is graduated N, E, S, W, N, E, S in the same manner as the wind direction portion of the chart is printed.

3.4.3.4 Wind speed component. The wind speed component shall contain an Esterline-Angus d.c. recording voltmeter or equivalent for providing a curvilinear movement to a pen to provide a trace of wind speed on the chart. The voltmeter shall be linearly calibrated so that 14.59 volts d.c. equals 120 knots, have an accuracy of ±1 per cent, an internal resistance of 6,900 ohms ±1/2 per cent and perceptible movement to a voltage change of 0.150 volts. An indicator scale shall be provided which is graduated in the same manner as the wind speed portion of the chart is printed. A permanently mounted 6,900 ohm shunt resistor, tapped at values of 1,380, 1,725, 2,300, and 3,450 ohms, shall be provided as a convenient means to maintain the correct load on the transmitter magneto as the number of repeaters in the circuit are varied.

3.4.3.5 Pens and ink well. A means of balancing the pens, lifting them from the chart, and zeroing them shall be provided. The pens shall draw lines 0.012 ±0.004 inches wide. The ink wells shall have at least a one (1) ounce capacity. Pens and ink wells shall be easily removable for cleaning.

3.4.3.6 Accessories. The following accessories shall be provided with each recorder:

a. Six rolls of charts.

b. One pint of red recorder ink.

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c. A detachable 8-foot a.c. power cable with necessary connectors.

d. An eight-circuit unwired plug connector for making connections from the wind direction-velocity transmitter to the recorder.

e. An accessory kit, containing one pen filler, one ink well filler, and two spare pen points.

3.4.4 Wind direction-velocity transmitter ML-400( )/UMQ-5. The transmitter shall consist of a streamline vane and a three-blade impeller mounted on a vane support. (See figure 1). Any movement of the vane shall be transmitted mechanically to a type 31TX6 synchro, MIL-S-20708, housed in the vane support. Any movement of the impeller shall be transmitted mechanically to a Weston Model 750, or equivalent, d.c. magneto housed in the vane. The vane shall turn the synchro transmitter to agree with the wind direction and shall keep the plane of the impeller normal to the air stream. The vane and vane support shall be of splash-proof construction per MIL-STD-108. The impeller blades shall be of molded plastic of type phenolic molding compound G.E. 12487 or equivalent. A plug and jack arrangement shall be provided to permit removal of the transmitter mechanism without removing the base connector from the Support MT-535( )/UMQ-5. A rain cap shall be provided to protect the jack when the transmitter mechanism is removed. The base connector shall contain a 3/4 inch pipe thread and plug for optional cable connection.

3.4.4.1 Size. The complete transmitter assembly shall not weigh more than 20 pounds and shall not exceed the dimensions shown on Figure 1.

3.4.4.2 Magneto output. The output generated by the d.c. magneto shall be linearly calibrated so that at 120 knots of wind speed the magneto shall supply 14.59 volts d.c. across a 1150 ohm load.

3.4.4.3 Stopping speed. The wind generator housed in its support with propeller attached shall have a stopping speed of not more than 2.4 knots.

3.4.4.4 Balance. The wind vane shall be statically balanced to within 5 ounce-inches of true static balance. The impeller shall be statically balanced to within 10 gram-inches.

3.4.4.5 Magneto. The magneto shall include the following features:

a. Housing shall be die cast aluminum alloy with dust tight enclosures.

b. Shaft shall be 3/8 inch diameter stainless steel.

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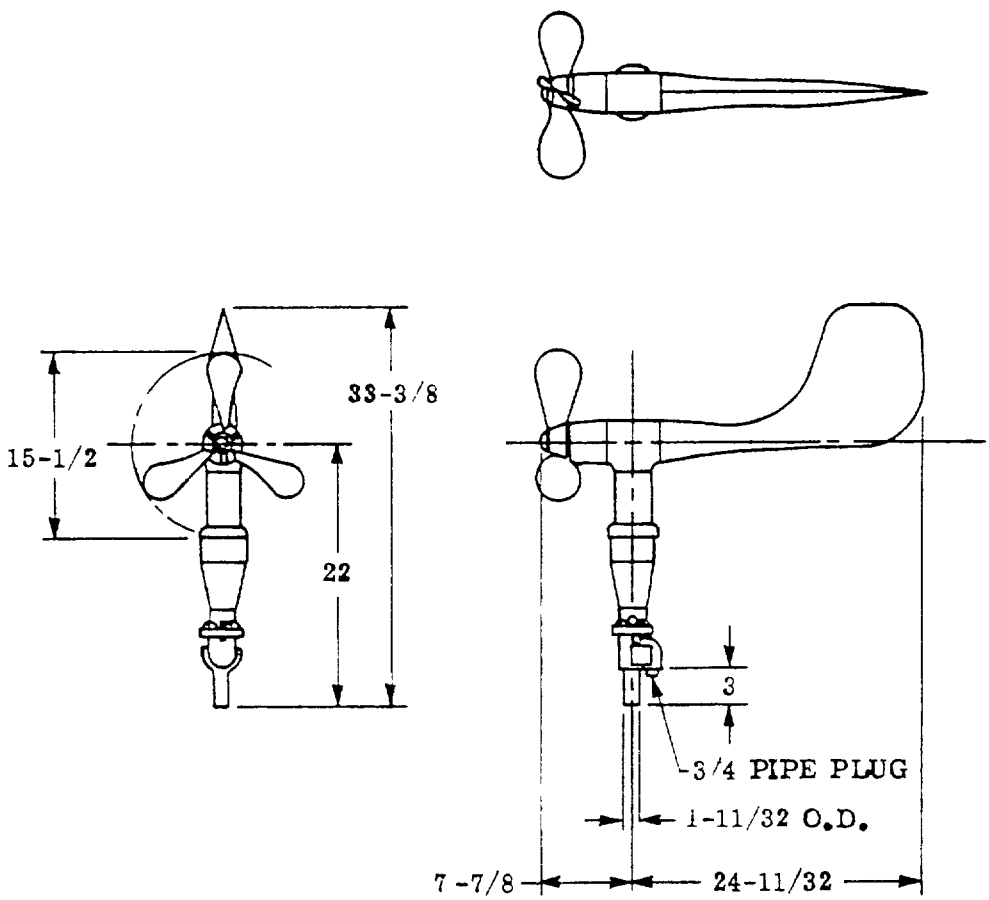


FIGURE 1. DIMENSIONS FOR TRANSMITTER ASSEMBLY

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c. Bearings - two shielded type ball bearings lubricated in accordance with MIL-G-23827.

d. Field magnet shall be ring-type cobalt steel with externally adjusted shunt.

e. Armature shall consist of at least 12 coils fed to lamina supported commutator having a minimum of 12 segments made of paladium silver alloy.

f. Brushes shall be finger type paladium silver mounted so as to equalize brush pressure.

g. Brush holder equipped with vibration dampener.

h. Brush assembly shall be removable and replaceable as a unit.

i. Ripple - the alternating current rms ripple shall not exceed 2% of the d.c. output voltage.

j. Resistance shall consist of 200 ohms -0.5% total, not over 15 ohms of which shall be copper, the balance to be magnanin for temperature compensation purposes.

3.4.4.6 Gaskets. Silicon rubber gaskets shall be installed between the upper and lower castings of the syncro housing on the transmitter assembly and the assembly belts.

3.4.4.7 Spare rotor. A spare rotor shall be furnished with each transmitter in addition to the equipment rotor. This spare rotor shall be included in the unit bid price for the transmitter.

3.4.5 Indicator, wind direction and speed ID-586( )/UMQ-5. The indicator shall contain, in an enclosure designed for desk use, two panel mounted dials, one a direction indicator, on the left side of the panel, incorporating a type 23TR6a Synchro, MIL-S-20708, for direction indications, and the other a speed indicator, on the right side, incorporating a General Electric Type DB-16 d.c. voltmeter. The mounting and interconnections of the speed and direction units shall be such as to facilitate the ready removal of the indicator panel for mounting in a standard 19-inch relay rack, and of either indicator for other separate, individual flush mounting, if desired.

3.4.5.1 Dials. The indicator dials shall be 8 ±1 inches in diameter, and shall be dull black with white printing and graduations. The wind direction dial shall be graduated in 5-degree intervals with identification numbers at each 30 degrees.

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Appropriate letters shall be used to designate the 4 cardinal and 4 inter-cardinal points of the compass. The wind speed dial shall be graduated from 0 to 120 knots in units of 2 knots, with distinctive marks at each 10 knots and with identification numbers at each 20 knots. The wind speed scale shall be graduated in 240 ±10 degrees of arc. White numerals and markings shall be "Semi-Gloss White 27875" per FED STD No. 595, or equivalent. There shall be no illumination provided.

3.4.5.2 Wind direction indicator. The front cover and dial of the direction indicator shall match that of the speed indicator (General Electric).

3.4.5.3 Wind speed indicator. The wind speed voltmeter shall be linearly calibrated so that 14.59 volts d.c. equals 120 knots, have an accuracy of ±1 per cent, an internal resistance of 6,900 ohms ±1/2 per cent, and perceptible movement to a voltage change of 0.090 volts. A permanently mounted 6,900 ohms shunt resistor, tapped at values of 1,380, 1,725, 2,300, and 3,450 ohms, shall be provided as a convenient means to maintain the correct load on the transmitter magneto as the number of repeaters in the circuit are varied.

3.5 Technical literature and spare parts. Technical literature and spare parts shall be furnished as specified by the contract. Spare parts shall be identical to corresponding parts in the equipment furnished on the order (See 6.2).

3.6 Workmanship. Components, subassemblies, and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Wind measuring sets covered by this specification shall be subjected to the following inspections.

a. First article inspection (see 4.3)

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b. Quality conformance inspection (see 4.4)

c. Inspection of preparation for delivery (see 4.5)

4.3 First article inspection. First article inspection shall be made on wind measuring sets representative of the production sets to be supplied under the contract. First article inspections shall be accomplished using the test procedures of 4.6. The government inspector and the procuring activity shall be advised when the inspections are to be conducted so that a representative may be designated to witness or supervise the inspections when so desired. Contractors not having adequate facilities to conduct all required inspections shall obtain the services of a commercial testing laboratory acceptable to the procuring activity.

4.3.1 First article inspection data. The contractor shall submit all data collected in conducting these inspections to the procuring activity for review and approval.

4.3.2 First article inspection samples. First article inspection samples shall consist of one complete Wind Measuring Set AN/UMQ-5( ).

4.4 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing all the inspections. Quality conformance inspection shall be under the supervision of the government quality control representative. Quality conformance inspection shall consist of the following tests:

a. Individual tests (see 4.4.1)

b. Sampling tests (see 4.4.2)

4.4.1 Individual tests. Individual tests shall be conducted on each wind measuring set submitted for acceptance and shall consist of the following test

Examination of product (see 4.7.1)

4.4.2 Sampling tests. The sampling plan shall be in accordance with MIL-STD-105 for inspection levels I or II. The acceptable quality level (AQL) shall be as follows

a. Major defects - 1.0

b. Minor defects - 10.0

The sampling tests shall consist of the tests in Table I.

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TABLE I

Inspection or Test	Paragraph
Examination of Product	4.7.1
Direction Indication	4.7.5
Speed Indication	4.7.6
Wind Direction-Speed-Transmitter Balance	4.6.10
Direction Vane Torque	4.7.11
Magneto Voltage	4.7.13
Wind Speed Component Torque Test	4.7.14

4.4.2.1 Major defects. Failure of equipment under any of the tests listed in paragraph 4.4.2 above, other than the examination of product test of 4.7.1, shall be classed as a Major Defect.

4.4.2.2 Minor defects. Individual defects noted during the examination of product test 4.7.1 shall in general be considered minor defects, but may be classed as major defects at the discretion of the cognizant inspector, if considered to seriously affect performance, durability or reliability of the equipment.

4.4.3 Disposition of defective items. The inspector shall determine whether or not a defective item may be corrected and re-submitted for acceptance and shall specify the conditions and circumstances under which re-testing and approval shall be accomplished.

4.5 Inspection of preparation for delivery. Samples from a quantity of shipping containers fully prepared for delivery, just prior to closure, shall be selected at random from each lot in accordance with MIL-STD-105 at Inspection Level S-2. The lot size shall be the number of shipping containers in the lot. Each sample selected shall be examined for defects and to determine conformance to all the applicable requirements of Section 5 of this specification. If the number of defective shipping containers exceeds the Acceptable Quality Level of 4.0, the lot represented by the sample shall be rejected.

4.6 Test conditions. Except for those tests where the following factors are the variables, tests shall be conducted with the equipment operating under the following conditions.

Temperature - the ambient temperature shall be 75°F ±20°F
Voltage - the supply voltage shall be 115 ±5 volts
Frequency - the supply frequency shall be 60 hertz

4.7 Test methods.

4.7.1 Examination of product. The completed equipment shall be

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given a thorough visual inspection to determine that it conforms to the applicable specifications and approved working drawings with respect to material, finish, workmanship, construction, assembly, dimensions, weight, identification marking and/or nameplates and instructions. This inspection shall be limited to those tests that may be performed without disassembling the component. This general inspection shall include a check of all operating controls, circuit functions, test facilities and adjustments as applicable and an operating test of the complete equipment as required in this specification.

4.7.2 Supply line voltage and frequency variation. The equipment shall perform satisfactory at line voltages of 115 ±10 per cent and at line frequencies of 55 to 60 hertz.

4.7.3 Interference. The Wind Measuring Set AN/UMQ-5( ) shall be tested to assure compliance with MIL-STD-461.

4.7.4 Continuous operation of transmitter. The wind speed and direction transmitter shall be capable of continuous operation as follows: The wind speed component shall be driven at a speed equivalent to 120 knots for a period of 24 hours. The direction component, including vane, shall be rotated at 5 RPM for a period of 24 hours. After this test, the transmitter shall be tested for accuracy of indications and examined for mechanical defects.

4.7.5 Direction indications. All indicators and recorders on the order shall be given this test. A pointer shall be attached to the shaft of a Type 23TR6a or Type 31TX6 synchro and shall operate over an azimuth scale plate at least eight inches in diameter. The synchro shall be connected electrically to the appropriate leads of the indicator or recorder and the system energized. The pointer shall be rotated continuously clockwise until the recorder pen or indicator pointer is successively positioned over each 30 degree interval. The test shall be repeated, rotating the pointer counterclockwise. No reading of the azimuth plate shall differ from the indicator and recorder readings by more than two (2) and four (4) geometric degrees, respectively.

4.7.6 Speed indications. All indicators and recorders on order shall be given this test. Variable direct current voltages shall be applied to the appropriate leads of the indicator or recorder so as to cause the recorder pen or indicator pointer to read 4, 10, 20, 40, 60, 80, 100, 120, 100, 80, 60, 40, 20, 10, and 4 knots. The true applied voltage for each of these pointer positions shall be read on a suitable voltmeter. The voltmeter readings shall be converted to speed values by multiplying the readings in volts by 8.226 (knots per volt). No computed speed shall differ from the indicated speed by more than ±1 knot over a range of 4 to 60 knots or by more than ±2 knots over a range of 61 to 120 knots.

4.7.7 Indicator and recorder response. The wind speed indicators and recorders shall be tested with a variable d.c. source and a suitable voltmeter

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for sensitivity of the speed meters to voltage change at 2, 4, 10, 20, 40, 80, 120 knots. At no point shall the voltage change required to produce perceptible movement of the indicators exceed 0.090 volts, and of the pen of the recorder exceed 0.150 volts.

4.7.8 Recorder chart drive test. The chart drive shall be operated at 6 inches per hour for a period of 48 hours. The drive shall have an accuracy of one minute per 24 hours from a 60 hertz source.

4.7.9 Recorder direction pen, repositioning time. The direction recorder unit shall be tested for time to reposition pen upon a reversal from near the end of the scale. The time elapsed from the instant reversal starts until the pointer comes to rest at its index position shall not exceed four (4) seconds. This test shall be performed for both directions.

4.7.10 Wind direction-velocity transmitter balance. The wind direction-velocity transmitter shall be mounted with its support horizontal. The transmitter shall be statically balanced to within five (5) ounce-inches of true static balance.

4.7.11 Direction vane torque test. The wind direction-velocity transmitter shall be mounted in the normal manner with its support vertical. Torque shall be gradually applied to the vane and the vane shall start rotating at two (2) ounce-inches of torque or less.

4.7.12 Transmitter accuracy. The wind direction-velocity transmitter together with an indicator and a recorder shall be tested in a wind tunnel to insure compliance with paragraph 3.3.4.

4.7.13 Magneto voltage. On production units, the following test shall be performed. The magneto shall be driven by a multi-speed motor at speeds listed in Table below. Voltages, across a 1,150 ohm load, shall be within ±0.054 volt of nominal value shown.

TABLE II

Rotor Shaft Speed R. P. M.	Nominal Voltage Output
400	2.044
1,200	6.133
1,800	9.200
2,400	12.266

4.7.14 Wind speed component torque test. With the wind direction-velocity transmitter mounted in the normal manner, the magneto shall start to turn at a torque of 1.0 ounce-inches or less.

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4.7.15 Stopping speed test. The wind speed transmitter shall be tested in a wind tunnel with the wind speed gradually decreasing until the impeller stops. Three readings of stopping speed shall be taken. The average stopping speed shall be not more than 2.4 knots.

4.7.16 Voltage output lag test. The wind direction-velocity transmitter shall be mounted in a wind tunnel in its normal mounting position. A brake and motor shall be attached to the transmitter to provide means to decrease or increase the speed of the transmitter from that of the wind in the tunnel. The wind tunnel shall be brought to a steady state, the speed of the transmitter changed by means of the brake or motor in steps, the brake or motor released and the output of the transmitter recorded against time. The test shall be conducted at each of the following steady state speeds of the wind tunnel.

a. 10 ±1 knots

b. 20 ±2 knots

c. 40 ±4 knots

At each wind tunnel speed the rotor speed shall be changed to each of the following speeds·

a. 0 knots - (rotor locked)

b. 33 ±2% and 67 ±2% overspeed and underspeed

c. 100 ±4% overspeed (at wind tunnel speeds of 10 & 20 knots only)

At each wind tunnel speed the time required for the voltage output to attain 63 per cent of the steady state value shall be determined for each rotor speed. The arithmetic mean of these times shall be taken and the number of linear feet of wind travel at the tunnel steady state speed for this average time value shall be computed. The linear feet of wind passage shall not exceed 16 linear feet.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning. All assemblies or parts of the AN/UMQ-5( ) shall be thoroughly cleaned in accordance with the best commercial practices.

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5.1.1.2 Packaging.

5.1.1.2.1 Transmitter ML-400( )/UMQ-5.

5.1.1.2.1.1 The transmitter shall be packaged Method IIB (container, barrier, container) per MIL-P-116 with desiccant. The transmitter shall be blocked and braced with pads or cells of corrugated fiberboard conforming to the requirements of PPP-B-636. Two (2) copies of Government-furnished Handbook of Operation and Maintenance Instructions pertaining to the equipment shall be packaged with each transmitter.

5.1.1.2.1.2 The equipment rotor and spare rotor shall be individually packaged Method III per MIL-P-116 for physical and mechanical protection.

5.1.1.2.2 Recorder RD-108( )/UMQ-5

5.1.1.2.2.1 The chart drive mechanism shall be removed from the recorder case and packaged Method IIB (container barrier container) per MIL-P-116 with desiccant. The mechanism shall be blocked and braced with pads or cells of corrugated fiberboard conforming to the requirements of PPP-B-636.

5.1.1.2.2.2 The recorder case shall be packaged in the same manner as the chart drive mechanism per paragraph 5.1.1.2.2.1.

5.1.1.2.2.3 Equipment Spares

5.1.1.2.2.3.1 The chart rolls in the vendor's box, and accessory kit shall be packaged per Method 1C3 of MIL-P-116 (waterproof bag, sealed).

5.1.1.2.2.3.2 The pens ink wells and bottle of ink shall be packaged per Method III of MIL-P-116.

5.1.1.2.2.3.3 The power cord connectors and cable plug shall be packaged per Method IA8 of MIL-P-116.

5.1.1.2.3 Wind Direction and Speed Indicator ID-300( )/UMQ-5A

5.1.1 2.3.1 The indicator shall be packaged Method IIB per MIL-P-116 with desiccant.

5.1.1.2.4 Wind Direction and Speed Indicator ID-586 ( )/UMQ-5

5.1.1.2.4.1 Method of packaging shall be the same as 5.1.1.2.3.1.

5.1.1.2.5 Support MT535( )/UMQ-5

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5.1.1.2.5.1 Support components shall be packaged per Method III of MIL-P-116.

5.1.2 Level B (See section 6)

5.1.2.1 Cleaning. All assemblies or parts of the AN/UMQ-5( ) shall be thoroughly cleaned in accordance with the best commercial practices.

5.1.2.2 Packaging. Transmitter ML-400, Recorder RD108, Indicator ID-300, Indicator ID-586, Support MT-535, shall be packaged in accordance with the requirements of paragraph 5.1.1.2 with the exception that no desiccant is required.

5.1.3 Level C

5.1.3.1 Cleaning. All assemblies or parts of the AN/UMQ-5( ) shall be thoroughly cleaned in accordance with the best commercial practices.

5.1.3.2 Packaging. Transmitter ML-400, Recorder RD108, Indicator ID-300, Indicator ID-586, Support MT-535, shall be packaged for physical protection only in accordance with Method III per MIL-P-116.

5.2 Packing

5.2.1 Level A

5.2.1.1 Transmitter ML-400( )/UMQ-5. Each packaged transmitter with packaged rotors and cable connector housing shall be packed in suitable export shipping containers conforming to the provisions of PPP-B-636.

5.2.1.2 Recorder RD-108( )/UMQ-5. The packaged components of the recorder and the packaged equipment spares shall be packed in suitable export shipping containers conforming to the provisions of PPP-B-636.

5.2.1.3 Wind Direction-velocity Indicator ID-300( )/UMQ-5. The packaged indicators shall be packed in suitable export shipping containers conforming to the provisions of PPP-B-636.

5.2.1.4 Wind Direction and Speed Indicator - ID-586( )/UMQ-5. Indicators shall be packed as per paragraph 5.2.1.3.

5.2.1.5 Support, MT535( )/UMQ-5. Packaged components of support shall be packed in a suitable shallow wooden box in accordance with applicable military specifications.

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5.2.2 Level B. Packaged transmitter ML-400, Recorder RD-108, Indicator ID-300, and Indicator ID-586 shall be individually packed in corrugated fiberboard cartons in accordance with specification PPP-B-636. Support MT-535 shall be packed in a suitable, shallow wooden box.

5.2.3 Level C. Packaged transmitter ML-400, Recorder RD-108, Indicator ID-300, and Indicator ID-586 shall be individually packed in corrugated fiberboard cartons in accordance with specification PPP-B-636. Support MT-535 shall be packed in a suitable, shallow wooden box.

5.3 Marking. All shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. The equipment shall be used to measure, indicate and record wind direction and speed.

6.2 Ordering Data. Procurement documents should specify the following

a. Title, number, and date of this specification.
b. Quantity of items required.
c. Technical literature required (3.5)
d. Quantity of spare parts required (3.5)
e. Selection of applicable levels of preservation and packaging, and packing.

(1) Level B preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined to be appropriate. Effective period of protection is 6 months from date of initial packaging.

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6.3 Availability of drawings. Bidders may obtain a copy of Naval Air Systems Command Drawing Set No. 58A43 from the procuring activity. This microfilm set will be furnished on loan upon request and submission of a deposit of $75.00. The deposit will be refunded upon return of the microfilm set. A complete microfilm set of drawings will be furnished to the contractor when selected, for use during the contract. The contractor shall apply to the Naval Air Systems Command, Attn: AIR-5403, Washington, D. C. 20360, for the set of drawings when required.

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Navy - AS
Project No. 6660-N010

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DOCUMENT IDENTIFIER AND TITLE MIL-W-17247B (AS) WIND MEASURING SET AN/UMQ-5 ( ) (COMPONENTS OF)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4 REMARKS

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	DATE

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